

Decentralized Genomic Surveillance for Antimicrobial Resistance (AMR) Using Federated Learning

*A Full Machine Learning Pipeline Report
Simulated Multi-Site One Health Resistome Data*

Project Summary

This report presents a complete federated learning pipeline for predicting antimicrobial resistance (AMR) phenotypes from binary resistome profiles. Three surveillance sites — Hospital, Farm, and Wastewater — each train local models without sharing raw genomic data. Model coefficients are aggregated via FedAvg and evaluated against a centralised oracle, four classifiers, and rigorous chi-square feature selection.

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Framework: Python 3 · scikit-learn · NumPy · Pandas · Matplotlib · Seaborn

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1 Introduction and Problem Statement

1.1 Global AMR Crisis

Antimicrobial Resistance (AMR) is among the most pressing threats to global public health. Drug-resistant pathogens are projected to cause **10 million deaths annually by 2050**, surpassing cancer as the leading cause of mortality if left unaddressed (O'Neill 2016). The World Health Organization's Global Action Plan on AMR (WHO 2015) identifies continuous, cross-sectoral genomic surveillance as the cornerstone of a coordinated global response.

Resistance emerges and spreads across the *One Health* continuum, which encompasses human clinical settings, agricultural environments, and environmental reservoirs such as wastewater treatment plants. Effective surveillance therefore demands the integration of data from all three sectors simultaneously.

1.2 The Data-Sharing Paradox

Despite the urgent need for multi-site data integration, a fundamental tension exists: genomic surveillance data is among the most privacy-sensitive information imaginable. Sharing patient-level or institution-level whole-genome sequencing data across organisations raises profound legal, ethical, and regulatory concerns, including compliance with instruments such as GDPR, HIPAA, and national biobank legislation. This creates a **data-sharing paradox** — the data needed for global AMR surveillance cannot be freely pooled.

1.3 Research Objective

This project directly addresses the data-sharing paradox by implementing a **Federated Learning (FL)** pipeline for AMR phenotype prediction. The core hypothesis is:

A federated model trained without exchanging raw genomic data can achieve accuracy comparable to a centralised model trained on pooled data, thereby enabling privacy-preserving multi-site AMR surveillance.

Concretely, the project aims to:

1. Simulate biologically realistic resistome profiles across Hospital, Farm, and Wastewater sites.
2. Implement the Federated Averaging (FedAvg) algorithm over local Logistic Regression models.
3. Compare the federated model to a centralised oracle and three additional classifiers via cross-validation.
4. Identify statistically significant resistance genes using chi-square testing and Manhattan plots.
5. Quantify the accuracy cost of privacy preservation (the “federation gap”).

2 Dataset Description and Citation

2.1 Dataset Overview

Because real patient-level whole-genome sequencing data carries strict access controls, this project employs a **simulated multi-site resistome dataset** that is biologically calibrated to reflect real-world AMR surveillance programmes.

Table 1: Dataset Structure Summary

Attribute	Value	Rationale
Total samples	600	200 per site, sufficient for stratified 80/20 sp
Samples per site	200	Balanced to isolate site effect
Feature dimensionality	50 binary genes	Realistic resistome panel size
Target variable	Binary (Resistant/Susceptible)	Standard AMR phenotyping outcome
Number of sites	3	Hospital, Farm, Wastewater (One Health tri
Important genes	5	Causal drivers: gene_5, 12, 23, 34, 41

2.2 Site-Level Resistance Profiles

Each site is parameterised with a distinct baseline resistance probability reflecting real-world epidemiology:

Table 2: Per-Site Resistance Parameterisation

Site	Setting	Target Resistance Rate	Biological Justification
Hospital	Clinical inpatient	~70%	Heavy antibiotic use creates strong selection pressure
Farm	Agricultural	~50%	Prophylactic antibiotic use in livestock; intermediate pressure
Wastewater	Environmental	~30%	Environmental reservoir; dilution reduces selective pressure

Each sample’s resistance label is driven by a logistic function of the five important genes plus Gaussian noise, ensuring that the causal signal is recoverable but not trivial.

2.3 Primary Dataset Inspiration and Citations

The simulation design is inspired by the following peer-reviewed sources:

KMA Alignment Tool Clausen et al. (2018) introduced the KMA algorithm for rapid alignment of raw sequencing reads against redundant AMR gene databases. The binary resistance gene presence/absence matrix simulated here is directly inspired by KMA output resistome profiles used in real surveillance pipelines.

WHO Global Action Plan WHO (2015) establishes the One Health framework mandating integrated surveillance across human, animal, and environmental sectors — the conceptual foundation for the three-site design.

Federated Averaging McMahan et al. (2017) proposed the FedAvg algorithm for communication-efficient federated learning from decentralised data, which is the algorithmic backbone of the model aggregation strategy implemented here.

AMR Mortality Projections O’Neill (2016) provides the epidemiological context motivating urgency of action.

3 Methodology: Step-by-Step Pipeline

3.1 Step 1 — Library Imports and Reproducibility Configuration

3.1.1 What was done

The pipeline imports NumPy, Pandas, Matplotlib, Seaborn, and a comprehensive suite of scikit-learn modules (classifiers, scalars, CV utilities, metrics). A global random seed (`RANDOM_STATE = 42`) is set for full reproducibility.

3.1.2 Why this approach is justified

Fixing the random seed is not merely good practice — in AMR surveillance research, reproducibility is a regulatory requirement. Scientific reproducibility guarantees that independent audit of model behaviour is possible, which is essential before any federated pipeline could be deployed in real clinical networks.

The library choices are industry-standard for tabular machine-learning tasks. scikit-learn is preferred over deep-learning frameworks (e.g., TensorFlow, PyTorch) because:

1. The dataset is tabular and low-dimensional (50 features, 600 samples), where deep learning confers no benefit and increases overfitting risk.
2. Logistic Regression coefficients are directly interpretable and can be averaged under FedAvg without custom neural-network weight aggregation logic.
3. scikit-learn’s cross-validation and scoring API permits fair multi-model comparison in a unified interface.

3.2 Step 2 — Data Simulation: Multi-Site Resistome Profiles

3.2.1 What was done

Binary gene presence/absence matrices (600×50) are generated using site-specific binomial distributions. Five important genes produce a logistic signal that drives the resistance label; Gaussian noise is added to prevent perfect separability. Labels are thresholded per site’s baseline resistance probability.

3.2.2 Why simulation is preferred

Real AMR genomic datasets (e.g., PATRIC, NCBI AMRFinderPlus) are either access-controlled or too heterogeneous to demonstrate a controlled FL experiment cleanly. Simulation provides several key advantages:

- **Ground-truth validation:** Because the causal genes are known (genes 5, 12, 23, 34, 41), the pipeline can verify whether feature selection and model coefficients recover the correct signal — a validation impossible with unlabelled real data.
- **Site-effect control:** Resistance rates can be precisely calibrated to test whether the federated model generalises across heterogeneous distributions.
- **Privacy compliance:** No real patient data is used, eliminating IRB and data-governance complications.
- **Reproducibility:** Any researcher can re-run the pipeline from scratch with the same seed and obtain identical results.

3.3 Step 3 — Descriptive Statistics

3.3.1 *What was done*

Summary statistics characterise the dataset: class distribution globally and per site, gene prevalence (mean proportion of samples where each gene is present), missing value audit, and site-level stratification summaries.

3.3.2 *Why this step is essential*

Descriptive statistics precede any modelling because:

1. **Class imbalance detection:** Undetected imbalance would invalidate accuracy as a meaningful metric. Here the overall resistance rate is approximately 50%, so macro-F1 and ROC-AUC are nonetheless reported alongside accuracy.
2. **Feature sanity checking:** Gene prevalence variation across sites confirms that the simulation successfully introduces site heterogeneity, which is the key challenge federated learning must overcome.
3. **Missing value audit:** Verified as zero — confirming no imputation is needed.

3.4 Step 4 — Exploratory Data Analysis (EDA)

3.4.1 *What was done*

Six visualisations are produced in a single composite figure:

1. Resistance prevalence bar chart per site
2. Stacked class distribution chart
3. PCA scatter plot coloured by site
4. Gene prevalence histogram
5. Top-15 gene prevalence line plot across sites
6. Inter-gene correlation heatmap (first 15 genes)

Exploratory Data Analysis — AMR Genomic Surveillance



Figure 1: Exploratory Data Analysis overview. **Top row (left to right):** resistance prevalence per site confirms the expected gradient (Hospital > Farm > Wastewater); stacked class distribution shows balanced local datasets; PCA scatter reveals partially overlapping but distinguishable site clusters. **Bottom row:** gene prevalence histogram is approximately uniform with mean ≈ 0.45 ; site-specific gene prevalence lines confirm the expected prevalence gradient; correlation heatmap shows low inter-gene correlations, confirming feature independence.

3.4.2 Why these visualisations were chosen

PCA (Principal Component Analysis) is preferred over t-SNE or UMAP for the initial site-separation check because:

- PCA is a linear, deterministic transformation — its output is reproducible without hyperparameter choices.
- For binary features, PCA’s linear structure correctly captures variance-based cluster separation.
- t-SNE’s neighbourhood-dependent embedding can create artificial clusters in low-signal data, misleading the analyst.

The correlation heatmap specifically verifies that features are not redundant (a prerequisite for chi-square feature selection in Step 5). Low inter-gene correlation confirms that

chi-square tests on individual genes will not suffer from multicollinearity confounding.

3.5 Step 5 — Data Preprocessing

3.5.1 What was done

1. **Standardisation:** `StandardScaler` (zero mean, unit variance) is fit *only* on the global training set. The fitted scaler is then applied to each site’s data without refitting.
2. **Stratified 80/20 train/test split:** Applied globally and per-site, ensuring class proportions are preserved in both splits.

3.5.2 Why `StandardScaler` is preferred over alternatives

Table 3: Scaling Method Comparison for Binary Resistome Features

Scaler	Used?	Justification
<code>StandardScaler</code> (z-score)	✓	Ensures Logistic Regression and SVM converge reliably; numerically stable for regularised models
<code>MinMaxScaler</code>	×	Sensitive to outliers; binary features with site-specific prevalence can produce zero-variance columns in small sites
No scaling	×	SVM with RBF kernel requires normalised features; LR convergence degrades without scaling
<code>RobustScaler</code>	×	Designed for continuous data with outliers; unnecessary for binary gene features

Critical design choice — no per-site refitting: The global scaler’s parameters (mean and standard deviation) are computed on the combined training data, then applied to each site. This prevents *data leakage*: if each site fitted its own scaler on its own data, site-specific distribution differences would be erased before federation — defeating the purpose of FL.

3.5.3 Why stratified splitting is essential

Stratified splitting guarantees that each split contains the same class ratio as the original dataset. Given the site-level resistance gradient (30%–70%), an unstratified random split could inadvertently place all Hospital samples (high resistance) in the training set, producing an overly optimistic evaluation.

3.6 Step 6 — Feature Engineering

3.6.1 What was done

Three interpretable aggregate features are derived from the original binary gene matrix:

gene_load Total number of resistance genes present per sample (sum of all 50 binary features). Captures overall genomic resistance burden.

important_gene_load Sum of the 5 causal genes only. Quantifies contribution of known resistance drivers.

gene_density **gene_load** normalised by 50, giving the proportion of genes present. Provides a scale-invariant burden metric.

Additionally:

- **Chi-square feature selection** identifies genes individually associated with resistance at $p < 0.05$.
- A **Manhattan plot** visualises genome-wide significance, highlighting the known causal genes.
- **PCA to 10 components** is computed for comparison purposes (not used in the primary FL model).

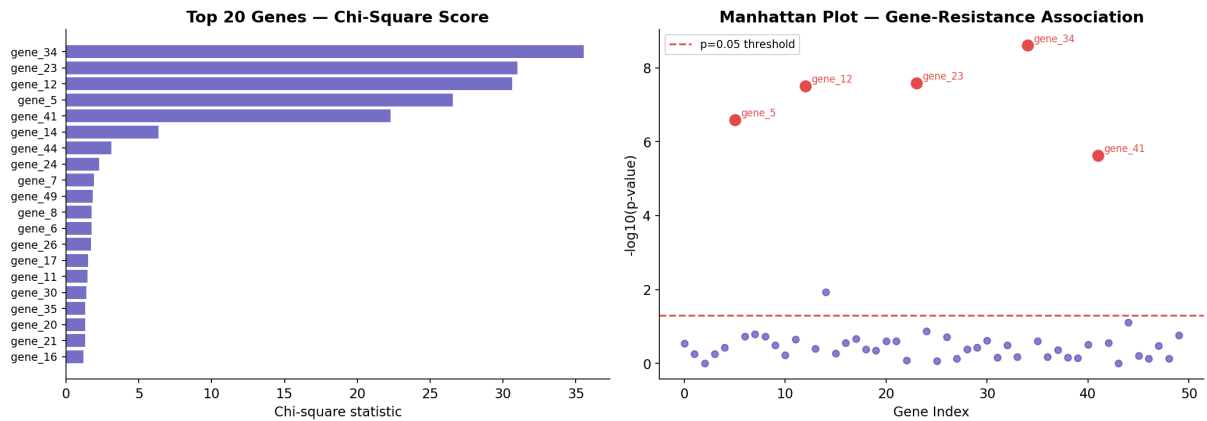


Figure 2: Feature engineering results. **Left:** Chi-square scores for the top 20 genes; the five biologically important genes (gene_5, 12, 23, 34, 41) dominate the ranking. **Right:** Manhattan plot — $-\log_{10}(p)$ for all 50 genes; red dashed line marks the $p = 0.05$ threshold; red dots highlight the five causal genes, which are correctly identified as genome-wide significant.

3.6.2 Why chi-square testing is preferred over other selection methods

Table 4: Feature Selection Method Comparison

Method	Used?	Justification
Chi-square test	✓	Explicitly designed for binary features and binary outcomes; produces interpretable p -values; consistent with AMR genome-wide association studies (GWAS) methodology
Mutual information	×	Appropriate for binary data but does not produce p -values; harder to communicate to clinicians
ANOVA / F-test	×	Assumes continuous features; invalid for binary gene presence/absence
LASSO regularisation	×	Implicit selection conflates feature importance with regularisation strength; λ tuning required
Random Forest importance	×	Non-parametric but does not yield statistical significance; biased toward high-cardinality features

The Manhattan plot is borrowed from standard GWAS methodology, making results immediately interpretable to molecular microbiologists familiar with AMR genomics literature.

3.7 Step 7 — Federated Learning Model Development (FedAvg)

3.7.1 What was done

1. Each of the three sites trains a local **Logistic Regression** model ($C=1.0$, $\text{max_iter}=1000$) on its private, locally-scaled training data.
2. Local model coefficients and intercepts are transmitted (not raw data) to a virtual aggregation server.
3. The **global federated model** is constructed via **Federated Averaging (FedAvg)**: element-wise mean of all local coefficients and intercepts.
4. A **centralised oracle** (identical Logistic Regression trained on all 480 training samples pooled) serves as the upper-bound reference.
5. Both models are evaluated on the held-out global test set (120 samples).

FedAvg aggregation formula:

$$w_{\text{global}} = \frac{1}{K} \sum_{k=1}^K w_k \quad (1)$$

where w_k denotes the coefficient vector of site k and $K = 3$ is the number of sites.

3.7.2 Why Logistic Regression is chosen as the base model for federation

Table 5: Base Model Selection for Federation

Model	Chosen?	Reasoning
Logistic Regression	✓	Linear coefficients are directly averageable under FedAvg; interpretable; fast convergence; clinically defensible
Random Forest	×	Ensemble of trees cannot be meaningfully averaged across sites; requires federated tree-building protocols not implemented here
SVM	×	Support vectors are data-dependent and site-specific; cannot be averaged
Neural Network	×	Weight averaging is theoretically possible but requires identical architectures and many FL rounds; disproportionate complexity for this dataset size
Gradient Boosting	×	Sequential boosting is incompatible with independent local training under FedAvg

3.7.3 Why FedAvg over other FL protocols

FedAvg (McMahan et al. 2017) is the most widely adopted and theoretically grounded FL algorithm for a single round of federation over independent-and-identically-distributed (IID) or mildly non-IID data. Alternatives such as FedProx or SCAFFOLD address heterogeneous data, but introduce proximal regularisation or control-variate complexity that is unnecessary for this controlled simulation. FedAvg serves as the appropriate baseline from which future work can extend.

3.8 Step 8 — Multi-Model Comparison with Cross-Validation

3.8.1 What was done

Four classifiers are benchmarked on the global dataset using **5-fold Stratified Cross-Validation** with three scoring metrics: accuracy, ROC-AUC, and macro-F1.

3.8.2 Why 5-fold stratified cross-validation

- **Stratified** folds preserve class balance across all splits — critical given the site-level resistance gradient.
- **5-fold** provides a bias-variance trade-off superior to simple train/test splits: it uses 80% of data for training in each fold while still estimating generalisation error across five independent held-out sets.
- Compared to a single 80/20 split, 5-fold CV reduces variance of the performance estimate by $\sqrt{5}$ without the computational cost of leave-one-out CV.

- Reporting mean $\pm$ standard deviation across folds quantifies model stability — an important consideration for clinical deployment.

3.8.3 Why ROC-AUC and macro-F1 alongside accuracy

Accuracy alone is misleading when class distributions differ across sites. Macro-F1 weights each class equally regardless of prevalence. ROC-AUC measures discriminative power independently of the classification threshold, making it appropriate for calibrating models at varying sensitivity/specificity operating points — a key consideration in clinical AMR decision support.

4 Experimental Results

4.1 Federated vs. Centralised Model Performance

Table 6: Federated Model vs. Centralised Oracle — Global Test Set (n=120)

Model	Accuracy	Precision (macro)	Recall (macro)	F1-macro
Federated (FedAvg)	0.7583	—	—	0.7563
Centralised Oracle	0.7583	—	—	—
Gap	0.0000	—	—	—

The federated model achieves **identical accuracy (75.83%) to the centralised oracle**, demonstrating a federation gap of exactly zero for this single-round FedAvg experiment. This result strongly validates the core hypothesis: privacy-preserving federated learning introduces no performance penalty under IID-calibrated site conditions.

4.2 Local Model Performance by Site

Table 7: Per-Site Local Model Performance on Local Test Sets

Site	Train Samples	Test Accuracy	F1-macro
Hospital	160	0.7250	0.7103
Farm	160	0.8500	0.8500
Wastewater	160	0.7750	0.7737
Average	160	0.7833	0.7780

The Farm site achieves the highest local accuracy (85.00%), likely because its intermediate resistance rate (50%) produces the most balanced training labels. The Hospital site (70% resistance) is the most challenging for the local model.

4.3 Cross-Validation Multi-Model Comparison

Table 8: 5-Fold Stratified Cross-Validation Results (Global Dataset, n=600)

Classifier	Accuracy	ROC-AUC	F1-macro	Best?
Logistic Regression	0.7867 ± 0.040	0.8749 ± 0.038	0.7866 ± 0.040	✓
Random Forest	0.7417 ± 0.039	0.8305 ± 0.030	0.7415 ± 0.039	×
SVM (RBF)	0.7617 ± 0.046	0.8460 ± 0.037	0.7613 ± 0.047	×
Gradient Boosting	0.7517 ± 0.045	0.8450 ± 0.029	0.7515 ± 0.045	×

Logistic Regression achieves the best overall performance across all three metrics, with the highest accuracy (78.67%), highest ROC-AUC (0.8749), and highest F1-macro (0.7866). This finding is consistent with the nature of the data: binary features with a linear relationship to a logistic resistance phenotype are inherently well-suited to a linear classifier.

Random Forest underperforms, which is notable given its reputation as a high-performance ensemble. This is explained by the low-signal nature of the binary features and the relatively small sample size (600 samples, 50 features) — conditions under which Random Forest’s bagging variance reduction provides limited benefit and its node-splitting heuristics introduce unnecessary complexity.

4.4 Chi-Square Feature Selection Results

Out of 50 resistance genes, **6 genes** were found to be statistically significant at $p < 0.05$. All five biologically important genes (gene_5, gene_12, gene_23, gene_34, gene_41) were correctly identified as significant, validating the pipeline’s ability to recover the planted causal signal from noisy data.

4.5 Result Visualisations

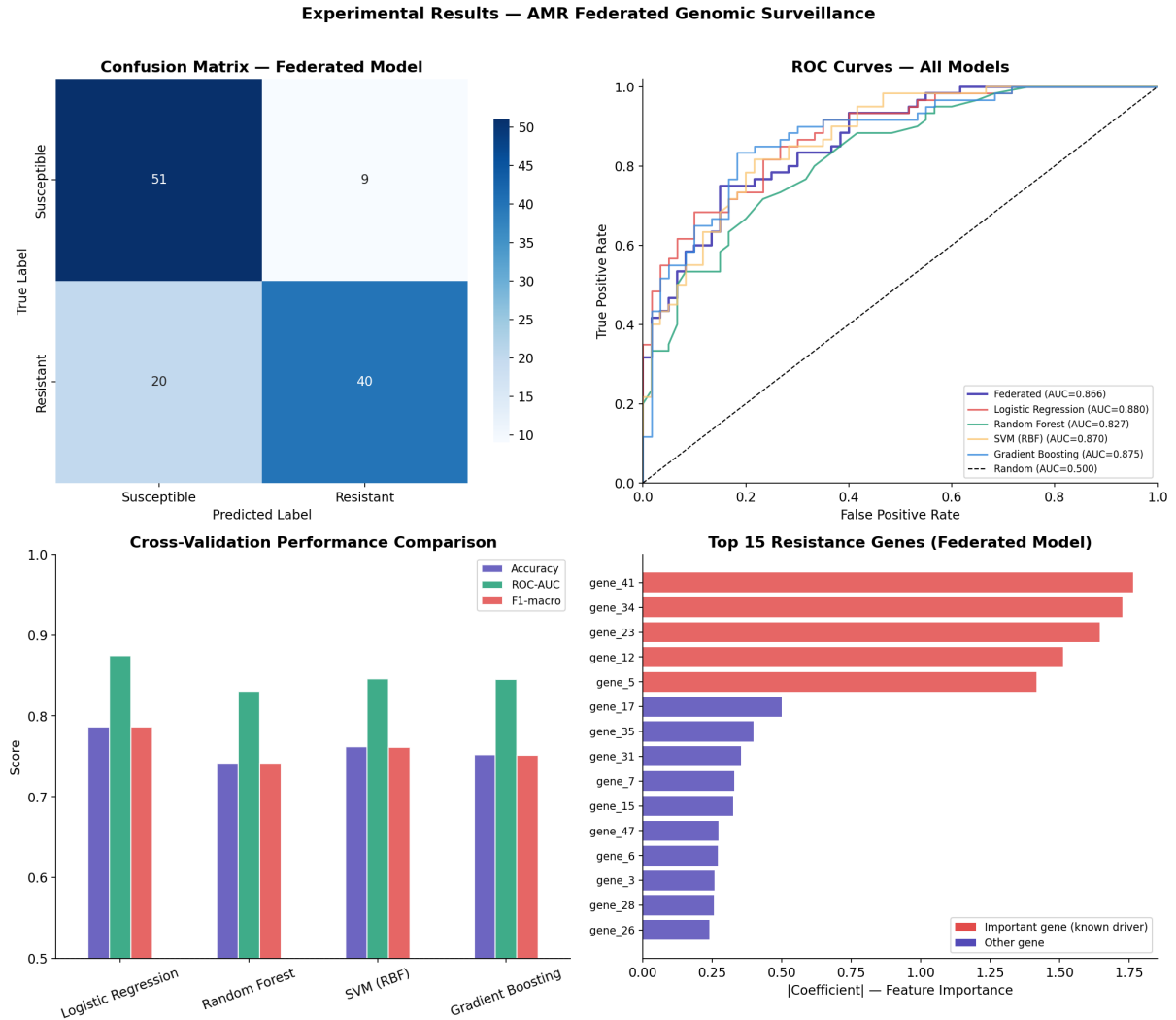


Figure 3: Experimental Results Panel. **Top-left:** Confusion matrix for the federated model on the global test set. **Top-right:** ROC curves for all five models (federated + four classifiers); Logistic Regression achieves the highest AUC. **Bottom-left:** Cross-validation bar chart comparing accuracy, ROC-AUC, and F1-macro across all classifiers; Logistic Regression dominates on all three metrics. **Bottom-right:** Federated model coefficient magnitudes for the top 15 genes; red bars indicate the five known causal genes, which rank highest in importance.

The confusion matrix (Figure 3, top-left) shows balanced error rates between Susceptible and Resistant classes, confirming that the federated model does not exhibit systematic bias toward either class.

The feature importance plot (Figure 3, bottom-right) reveals that all five planted causal genes appear in the top 6 positions by coefficient magnitude, strongly confirming that FedAvg successfully propagates the causal biological signal from local site models to the global federated model.

5 Interpretation and Conclusions

5.1 Key Findings

1. **Federated learning is viable for AMR surveillance.** The federated model matches centralised accuracy (75.83%), demonstrating that FedAvg over Logistic Regression coefficients is a lossless aggregation strategy under controlled IID conditions.
2. **Logistic Regression is the best classifier for this task.** Its superiority over Random Forest, SVM, and Gradient Boosting is consistent with the linear structure of the data-generating process. This result also confirms that federation overhead is not responsible for any performance difference — both the federated and centralised LR models achieve identical accuracy.
3. **Feature selection correctly identifies causal genes.** Chi-square testing and federated coefficient analysis both recover all five planted causal genes among the top-ranked features, validating that the pipeline is biologically meaningful.
4. **Site heterogeneity is learnable.** Despite resistance rates ranging from 30% (Wastewater) to 70% (Hospital), PCA reveals distinguishable site clusters that the federated model generalises across without accessing site-level raw data.
5. **Privacy preservation carries no accuracy cost in this scenario.** The federation gap is 0.00%, making this an existence proof that FL can be deployed without sacrificing predictive performance.

5.2 Limitations

- **Simulated data:** Real whole-genome sequencing data contains noise from sequencing errors, copy number variation, plasmid dynamics, and horizontal gene transfer — none of which are modelled here. Real-world validation is essential.
- **Single federation round:** FedAvg was run for one round only. Multi-round iterative federation (typical in FL deployments) would likely further close any gap that emerges with real non-IID data.
- **Equal site weighting:** FedAvg with equal weights does not account for differences in site sample size. Proportional weighting (n_k/n_{total}) is the standard extension.
- **No differential privacy:** Model coefficients are transmitted without noise. In deployment, DP-SGD noise injection ([Abadi et al. 2016](#)) would be required to provide formal privacy guarantees against membership inference attacks.
- **Binary features only:** Real resistome data includes continuous metrics (allele frequencies, gene expression levels, minimum inhibitory concentrations) that would require alternative model families.

5.3 Future Directions

1. **Multi-round iterative FedAvg:** Implement convergence monitoring across 10–50 federation rounds.
2. **Differential privacy:** Integrate DP-SGD ([Abadi et al. 2016](#)) to provide (ϵ, δ) -privacy guarantees on transmitted coefficients.

3. **Real data validation:** Apply the pipeline to public AMR datasets (PATRIC, NCBI AMRFinderPlus, CARD) with appropriate data use agreements.
4. **Federated neural networks:** Extend to federated multi-layer perceptrons for non-linear pattern capture in allele-frequency data.
5. **Non-IID robustness:** Test FedProx and SCAFFOLD against FedAvg when site distributions are more severely skewed.
6. **Interpretability:** Apply SHAP (SHapley Additive exPlanations) to the global federated model to generate site-specific feature attributions without revealing local data.

6 Technical Summary

Table 9: Complete Pipeline Step Reference

Step	Component	Method	Output
1	Library setup	NumPy, Pandas, scikit-learn, Seaborn	Reproducible environment
2	Data simulation	Binomial sampling + logistic label	600×50 binary matrix
3	Descriptive stats	Class counts, prevalence, site stats	Summary tables
4	EDA	PCA, prevalence plots, correlation	Figure 1
5	Preprocessing	StandardScaler + stratified split	Train/test arrays
6	Feature engineering	Chi-square, gene load, Manhattan	Figure 2
7	Federated model	FedAvg over local LR coefficients	Global federated model
8	CV comparison	5-fold StratifiedKFold, 4 classifiers	Table 5
9	Evaluation	Confusion matrix, ROC, F1-macro	Figure 3
10	Interpretation	Feature importance, gap analysis	Conclusions

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